Pixels, Frames & Transformers

Detection, Segmentation and Counting · Module 12

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What Is in the Image, and Where?

Detection
boxes around objects, each with a category and a confidence score

Segmentation
the exact outline of an object, pixel by pixel

Counting
how many of something are visible

They let you code *parts* of an image: how many people are at a rally, whether police appear, how much of a cereal box a mascot takes up.

Standard Detectors Know 80 Things

RT-DETR, a transformer-based detector, is trained on **COCO**: 80 everyday categories such as *person*, *car*, *umbrella*, *backpack*.

No *police officer*, no *banner*, no *flag*. Like ImageNet's categories (module 06), they were chosen for a benchmark, not for your question.

The confidence threshold
Every box has a score. A low threshold finds more objects and more false ones; a high threshold misses partly hidden ones. Keep all boxes above a low score once, then try thresholds without rerunning the model.

Counting People Is Hard

- People hide behind each other
- Distant heads are a few pixels tall
- Detectors return a limited number of boxes
- The frame shows one moment and one part of the event

Validation: your codebook says *crowd* = ten or more visible people, so "10+ detected" is a rule fixed in advance that can be scored on the test set.

A count is a lower bound on the people visible and detectable in one photo, not the size of a demonstration.

Crowd size estimation uses area times density, often from several or aerial images, and reports a range.

Open-Vocabulary Detection

OWLv2 scores every candidate box against **text queries** with a CLIP-style text encoder (module 09):

```
"a police officer", "a protest sign", "a banner", "a flag"
```

Image-level codes: the highest score per query, above a threshold, validated against your *police* and *signs* codes.

Threshold and wording both matter. "police", "a police officer in uniform" and "a person wearing a police uniform" give different results.

Choose them on a separate hand-coded validation sample, then score the test set once.

Segmentation with SAM 2

SAM 2 (Meta, Apache 2.0) outlines an object when you point at it with a click or a box. It doesn't know *what* it outlines, so we combine it:

1. OWLv2 finds "a cartoon character"
2. SAM 2 traces it inside the box
3. measure: the share of the package front covered



here the box is almost three times the area of the mask

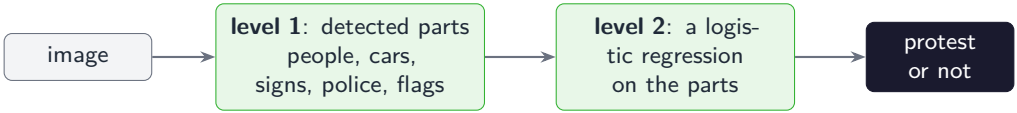
Kirillov et al. (2023) introduced Segment Anything; SAM 2 extends it to video (Ravi et al., 2024).

Measures Built from Regions

- **count**: people, flags, police vehicles
- **size**: share of the image a mascot or a speaker takes up
- **position**: foreground or background, center or edge
- **overlap**: is the sign held by a person?

Before you claim that mascots are used to sell less healthy cereals, validate the detector against hand codes, check that a few brands with many variants don't drive the pattern, and think about confounders such as the target age group.

Two-Level Classification



Readable: the coefficients say which objects push an image toward "protest", and expose shortcuts (a large weight on *umbrella* says something about the weather in your training data). **The price**: whatever the parts don't capture (setting, mood) is lost. Adding CLIP embeddings usually scores higher and reads worse.

Scholz, Weidmann, Steinert-Threlkeld, Keremoglu & Goldlücke (2025), *Political Analysis*.

Licenses: Check the Code and the Weights

Many detection tutorials use YOLO models from the Ultralytics library. Its recent versions are licensed under **AGPL-3.0**: software built with it must be released under the same license, also when offered as a web service, unless you buy a commercial license.

In this module: RT-DETR, OWLv2 and SAM 2 are Apache 2.0, which asks only for attribution and a copy of the license.

Fine for a published research script either way, but it matters for tools you build for partners. Code and weights can have different licenses.

In the Notebook

- Detecting and counting people with RT-DETR, checked against your crowd codes
- Finding police and signs with OWLv2, and how thresholds and wording matter
- Outlining cereal mascots with SAM 2 and measuring how much of the box they cover
- A two-level classifier from detected objects, and what it costs in accuracy

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12_detection_segmentation_and_counting_exercises.ipynb · 5 exercises · GPU needed